

Tutorial 7

1. A social scientist studies adult South Africans living in townships on the outskirts of Cape Town, to investigate educational attainment in the black community. Educational attainment is the number of years of education completed. Many of the study's potential subjects were forced to leave Cape Town in 1966 when the government passed a law forbidding blacks to live in the inner cities. Under the apartheid system, black South African children were not required to attend school, so some residents had very little education.

How large a sample size is needed so that a 95% confidence interval for the mean number of years of attained education has a margin of error of at most 1 year?

2. Recent research suggests that dogs may be helpful in detecting when a person has cancer. Six dog of varying breeds were trained to distinguish a patient with bladder cancer by smelling certain compounds in the patient's urine¹. The dog were trained to indicate which among several specimens was from the bladder cancer patient by lying beside it.

Each of the six dogs were tested with 9 trials. In each trial, one urine sample from a bladder cancer patient was randomly placed among six control urine samples. In the total of 54 trials with the six dogs, the dogs made the correct selection 22 times.

Let p denote the probability that a dog makes the correct selection on a given trial. Perform a significance test to test if the dog's ability to detect the correct urine specimen is better than random guessing.

- (a) State the assumptions for the test.
 - (b) State the hypotheses of the test.
 - (c) Derive the test statistic and its null distribution.
 - (d) Derive the p-value of the test. What is your conclusion?
3. A recent General Social Survey (in US) asked respondents, "On the average day, about how many hours of television do you personally watch?" Randomly collected data on the hours of TV watching (variable "hours") are given in the file `TVhours_long.csv`.

Let μ denote the average hours of TV watching in the population.

- (a) Report the sample size, sample mean and sample standard deviation.
- (b) Plot a histogram of the hours of TV watching. Give your comment on whether the sample is normally distributed.
- (c) Derive a point estimate of μ .
- (d) Perform a test (with 5 steps) at significance level 0.05 to test

$$H_0 : \mu = 4 \quad \text{vs} \quad H_1 : \mu \neq 4.$$

- (e) What is the margin of error at the 95% confidence level for the point estimate of μ ? Calculate a 95% confidence interval of μ . Interpret the obtained interval.

¹Article by C. M. Willis et al., *British Medical Journal*, vol. 329, September 25, 2004